

Akshatha Dungi

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EXPERIENCE

IDEAS- Institute of Data Engineering, Analytics and Science Foundation

Jun 2025 – Aug 2025

Summer Research Intern

ISI Kolkata

- Engineered an Artificial Neural Network (ANN) achieving 97% multi-class classification accuracy, improving predictive performance by 15% over existing models
- Designed and implemented model monitoring pipeline tracking accuracy, loss, and drift metrics across training runs, enabling faster iteration cycles
- Technical Stack: **Python, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib**

PROJECTS

AI Confidence-Calibration for Financial Time Series

- ★ Led a 5-person research team to systematically evaluate calibration failure across 5 neural architectures (**MLP, Deep MLP, Vanilla RNN, LSTM, Residual MLP**) on **15 years of financial news stock data**
- ★ Implemented Expected Calibration Error (**ECE**) tracking to measure confidence-accuracy misalignment over sequence lengths up to 100 timesteps
- ★ Discovered that Vanilla RNNs fail earliest (**t=25**) but most predictably (**r=0.89 correlation with gradient norms**), while **LSTMs mask failures until sudden collapse (t=75, r=0.34)**
- ★ Achieved **3.34% ECE** benchmark using Focal Calibration Loss, matching state-of-the-art results from **FinSen research**
- ★ **Built unified data pipeline processing 160K records** with custom loaders for all 5 models
- ★ Published complete codebase with reproducibility guidelines on **GitHub**

LEADERSHIP AND MANAGEMENT

a-hub, Incubation center

Jan 2023 – May 2023

Intern

Visakhapatnam

- Co-organized a campus internship drive from scratch sourced and onboarded 10+ corporate partners, coordinated scheduling across Business, Engineering, and IT tracks, and directly facilitated 30+ student placements in a single event cycle.

Andhra University College of Engineering

Nov 2022 – May 2026

Class Monitor & Research Team Lead

Visakhapatnam

- Served as Class Monitor for 2 years, acting as primary liaison between 70 students and faculty coordinating schedules, resolving conflicts, and ensuring smooth communication across a 4-year engineering cohort.
- Led a 5-person cross-functional research team on a 6-month AI calibration study, coordinating model implementation, data pipeline development, and results publication across the group.

EDUCATION

Andhra University College of Engineering, Visakhapatnam

Oct 2022 – Apr 2026

Bachelor of Technology in Computer Science and Systems Engineering

CGPA: 8.96/10

- ★ **Achievement:** Academic Excellence Award – Highest CGPA in CSE Department (2026)
- ★ **Coursework:** Data Structures, Algorithms, Databases, Computer Networks, Operating Systems, Object-Oriented Programming, Machine Learning

TECHNICAL SKILLS

Frontend : HTML5, CSS3, React, Responsive Design, JavaScript, Tailwind CSS, Bootstrap

Backend : Nodejs, Express, EJS, RESTful APIs

Programming Languages : C++, C, Python, Java

AI/ML Development : Python ML Stack(**scikit-learn, TensorFlow, PyTorch**), LLM Integration (OpenAI API, LangChain), AI-Assisted Development (GitHub Copilot), Neural Networks(**RNN, LSTM, MLP, Deep MLP, Residual Networks**), **Model**

Calibration, Expected Calibration Error (ECE), Reliability Diagrams, Time Series Analysis Forecasting

Databases : MySQL, PostgreSQL, MongoDB

Cloud/DevOps : Microsoft Azure, Git, GitHub Actions, CI/CD Pipelines, Docker